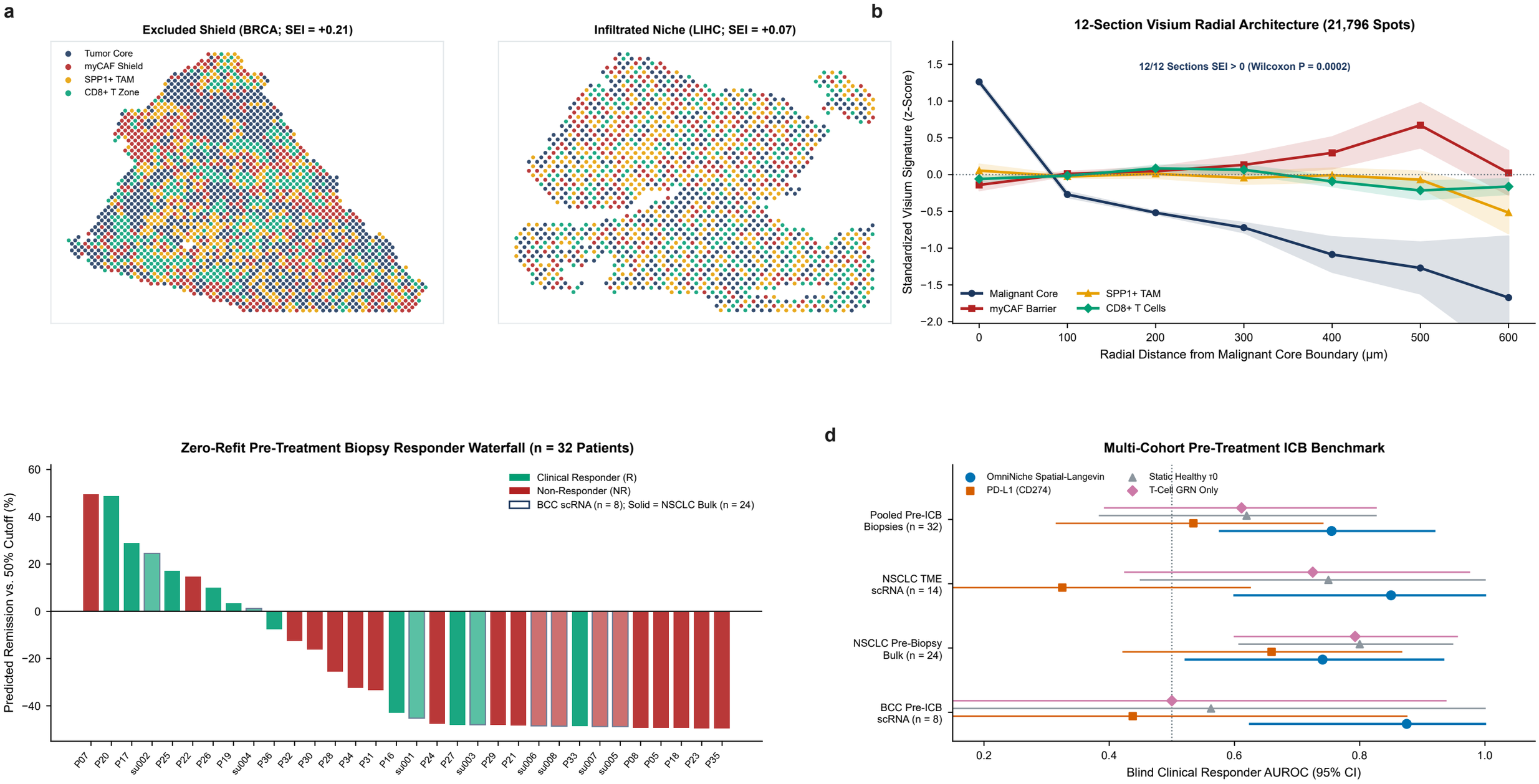

a
Excluded Shield (BRCA; SEI = +0.21)
Tumor Core
myCAF Shield
SPP1+ TAM
CD8+ T Zone
Infiltrated Niche (LIHC; SEI = +0.07)
b
12-Section Visium Radial Architecture (21,796 Spots)
12/12 Sections SEI > 0 (Wilcoxon P = 0.0002)
Standardized Visium Signature (z-Score)
Malignant Core
myCAF Barrier
SPP1+ TAM
CD8+ T Cells
Radial Distance from Malignant Core Boundary (μm)
c
Zero-Refit Pre-Treatment Biopsy Responder Waterfall (n = 32 Patients)
Clinical Responder (R)
Non-Responder (NR)
BCC scRNA (n = 8); Solid = NSCLC Bulk (n = 24)
Predicted Remission vs. 50% Cutoff (%)
P07 P20 P17 su002 P25 P22 P26 P19 su004 P36 P32 P30 P28 P34 P31 P16 su001 P24 P27 su003 P29 P21 su006 su008 P33 su007 su005 P08 P05 P18 P23 P35
d
Multi-Cohort Pre-Treatment ICB Benchmark
OmniNiche Spatial-Langevin
PD-L1 (CD274)
Static Healthy τ0
T-Cell GRN Only
Pooled Pre-ICB Biopsies (n = 32)
NSCLC TME scRNA (n = 14)
NSCLC Pre-Biopsy Bulk (n = 24)
BCC Pre-ICB scRNA (n = 8)
Blind Clinical Responder AUROC (95% CI)
e
Paired Pre/Post Anti-PD-1 Biopsies (n = 15)
Responders (n = 8)
Non-Responders (n = 7)
Net CD8+ Cytotoxic Program Score
Pre (R)
Post (R)
Pre (NR)
Post (NR)
f
Barrier-Unlocked 2D Attractor (PD-1 + TGFBR1)
myCAF Lock
Unlocked Remission
Malignant Burden (S_Tumor)
myCAF Barrier Activity (S_myCAF)
U_combo
g
In Silico Rescue in Resistant Patients (n = 19)
Mean 16.3% → 99.2%
(Wilcoxon P = 1.9e-6)
Simulated Remission Probability (%)
Anti-PD-1 Monotherapy
Anti-PD-1 + myCAF/TGFBR1